18 EXAM PRACTICE

1 Benzene and cyclohexene are tested separately with bromine water. Which observations are made?

A benzene and cyclohexene both decolourise bromine water

B benzene decolourises bromine water, but cyclohexene does not

C neither benzene nor cyclohexene decolourise bromine water

D benzene does not decolourise bromine water, but cyclohexene does [1]

(Total for Question 1 = 1 mark)

2 Bromine reacts with benzene on heating in the presence of a catalyst. It reacts with cyclohexa-1,3-diene at room temperature. What types of reactions are involved?

	bromine with benzene 	bromine with cyclohexa-1,3-diene 
A	addition	addition
B	addition	substitution
C	substitution	addition
D	substitution	substitution

[1]

(Total for Question 2 = 1 mark)

3 Benzene reacts very slowly with chlorine. Why does the reaction speed up when finely divided iron is added?

A the chlorine molecule donates an electron pair to the iron producing an electrophile

B the iron reacts with chlorine to form iron(III) chloride, which then acts as an electrophile in its reaction with benzene

C the iron reacts with chlorine to form iron(III) chloride, which then acts as an electron pair acceptor

D iron is a transition metal and acts as a heterogeneous catalyst in the reaction [1]

(Total for Question 3 = 1 mark)

4 It is calculated that benzene has a stabilisation energy of 150 kJ mol^{-1}, as a result of delocalisation. What does this mean?

A the enthalpy change for the conversion of benzene to cyclohexa-1,3,5-triene is +150 kJ mol^{-1}

B the enthalpy change for the conversion of cyclohexa-1,3,5-triene to benzene is +150 kJ mol^{-1}

C the enthalpy change for the conversion of cyclohexene to benzene is +150 kJ mol^{-1}

D the enthalpy change for the conversion of benzene to cyclohexene is +150 kJ mol^{-1} [1]

(Total for Question 4 = 1 mark)

5 Benzene reacts with a nitrating mixture of concentrated nitric and sulfuric acids. Which species is least likely to be present in the nitrating mixture?

A NO_3^-

B H_3O^+

C HSO_4^-

D NO_2^+ [1]

(Total for Question 5 = 1 mark)

6 The reaction between hydrogen and benzene is normally very slow. Why does it speed up in the presence of finely divided nickel?

A the nickel acts as a homogeneous catalyst by acting as an electron pair acceptor in the formation of an electrophile

B the nickel acts as a homogeneous catalyst by acting as an electron pair donor in the formation of an electrophile

C the nickel acts as a heterogeneous catalyst by providing active sites on which reaction occurs

D the nickel reacts with the hydrogen to form nickel(II) hydride, which then reacts with the benzene [1]

(Total for Question 6 = 1 mark)

7 Addition and substitution reactions are common types of reactions in organic chemistry.

(a) (i) Bromine reacts with both propene and benzene. Write an equation for each of these reactions. [2]

(ii) State the type of mechanism in each of these

reactions. [2]

(b) The reagent used in the nitration of benzene is concentrated nitric acid. The reaction is usually carried out using a catalyst of concentrated sulfuric acid.

Write equations to show how the sulfuric acid acts as a catalyst. [2]

(Total for Question 7 = 6 marks)

8 A Kekulé structure of benzene suggests that the molecule consists of alternate single and double carbon to carbon bonds.



The standard enthalpy change of hydrogenation of a carbon to carbon double bond is -120 kJ mol^{-1}.

(a) (i) Calculate the standard enthalpy change of hydrogenation of benzene for the Kekulé structure. [1]

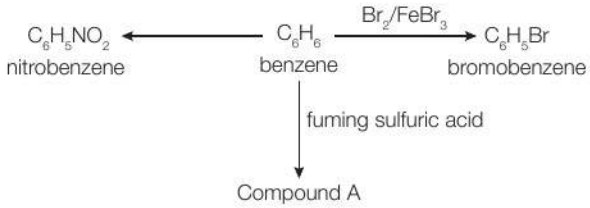
(ii) The actual standard enthalpy change of hydrogenation of benzene is -208 kJ mol^{-1}.

Use this information, and your answer to (i), to calculate the difference in stability of benzene and the Kekulé structure.

State what this tells us about the bonding in benzene.

Explain how this influences the type of chemical reactions that benzene undergoes. [4]

(b) The flow chart shows some typical reactions of benzene.



(i) Name the two chemicals needed to make nitrobenzene from benzene. [2]

(ii) Classify the type of reagent required in all three reactions. [1]

(iii) Show, with appropriate diagrams, the mechanism for the reaction of benzene with bromine, in the presence of iron(III) bromide, to form bromobenzene. Include an equation to show formation of the ion attacking the benzene molecule. [4]

(iv) Give the structural formula and name of compound A. [2]

(Total for Question 8 = 14 marks)

9 Benzene reacts with a mixture of concentrated nitric acid and concentrated sulfuric acid to form nitrobenzene.

(a) Give the mechanism for this reaction, including an equation for the formation of the electrophile. [4]

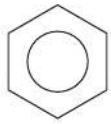
(b) The preparation of nitrobenzene is usually carried out between 50 and 60 °C.

Explain why the temperature used should not be higher or lower. [2]

(Total for Question 9 = 6 marks)

10 Benzene can be represented by either structure A or structure B.


structure A


structure B

Explain how the evidence from each of the following supports the view that structure B is the better representation for benzene:

(a) X-ray diffraction [2]

(b) enthalpy change of hydrogenation [2]

(c) the type of reaction that occurs with bromine. [2]

(Total for Question 10 = 6 marks)